

Sri Lanka Institute of Information Technology



IE1044 – Digital Electronics

Assignment – Design a Mini processor for BCD Arithmetic
Operations

1st year 2nd semester - 2025

IT24103479

Withanage M.P.A

Content

1. Introduction	3
2. Objectives	4
3. System Overview	5
a. Input Registers	5
b. BCD Adder	6
c. BCD Subtractor	6
d. BCD Multiplier	6
e. BCD Multiplexer	6
f. Output Registers	6
g. Display System	6
4. Components and Description	7
a. Mini Processor	7
b. BCD Adder	8
c. BCD Subtractor	9
d. BCD Multiplier	10
e. Comparator	11
f. Binary to BCD Converter	11
5. Simulation Results	12
6. Challenges and Fixes	20
7. Conclusion	21
8. References	22

1. Introduction

Binary – Coded Decimal (BCD) is a set of binary representation of decimal numbers in which every digit of the number is represented by an independent binary string. BCD arithmetic is used in digital equipment like calculators, digital clocks, and digital meters where accuracy in the decimal digits is of greatest significance.

The goal of this project is to develop a BCD arithmetic unit for addition, subtraction, and multiplication using modular blocks. Multiplexer is utilized to select desired operation, and registers are utilized to store inputs and outputs. The output is in terms of BCD to 7-segment display logic.

2. Objectives

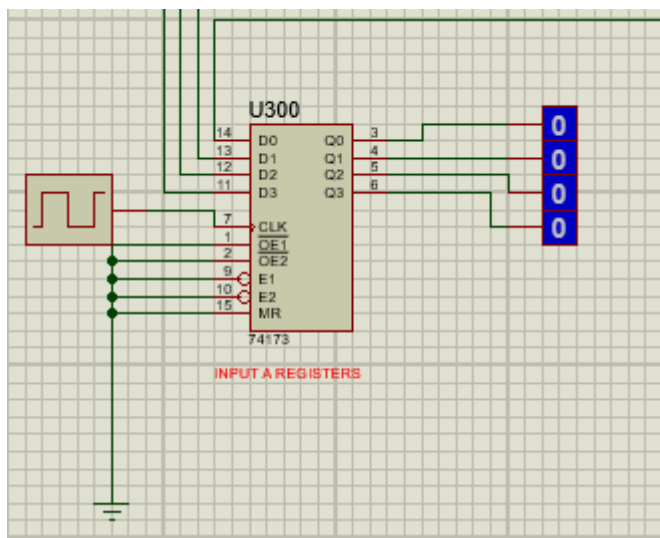
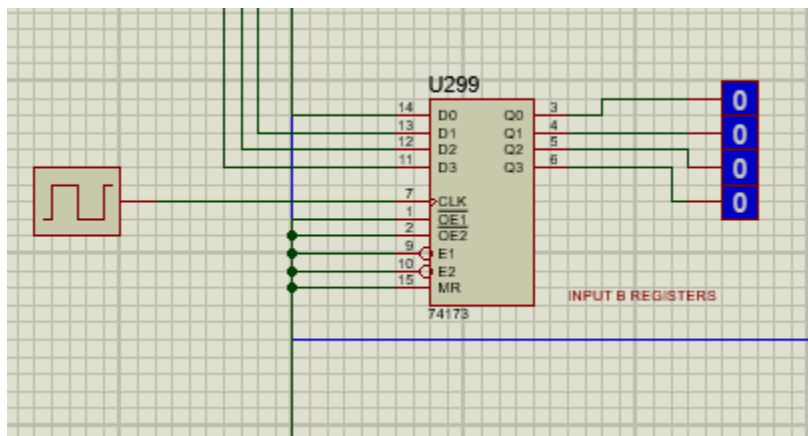
1. Develop a BCD Arithmetic unit to add, subtract, and multiply in modular form utilizing digital modules.
2. Implement individual logic modules (BCD adder, BCD subtractor, BCD multiplier) using TTL standard ICs and logic gates.
3. Dynamically change arithmetic operation via multiplexers.
4. Utilize registers to store inputs and results and clock control for synchronization.
5. Convert resulting binary to BCD as needed and output on 7-segment displays.
6. Detect and flag overflow conditions when the output is out of the valid range of BCD (especially during multiplication).
7. Simulate and test the whole system in Proteus with at least five working test cases.

3. System overview

The BCD arithmetic unit is designed as a mini processor composed of the following major components:

a. Input registers

Two 74173 4-bit registers are used to store the BCD inputs A and B. Each register has a clock and load enable input, allowing the user to load new values using a control signal. This approach prevents glitches and ensures stable operand storage before performing any operation.



b. BCD Adder

7483 binary adder with extra correction logic. When the sum is more than 9 (1001), it adds 6 (0110) to produce valid BCD output

c. BCD Subtractor

A 7483 adder is used here too, utilizing two's complement logic by complementing the B input with XOR gates and adding 1 through the carry-in. The circuit has correction logic and borrow detection through the carry-out. A second AND gate is used in combination with an A

selector to ensure that the minus sign is only displayed when the result is negative, not zero (e.g., not to display "-0" for $0 - 0$).

d. BCD Multiplier

A 7-bit output BCD digit multiplier was implemented based on a circuit borrowed from ResearchGate. The digit-wise multiplication ($0-9 \times 0-9$) is carried out by the multiplier, and later its 7-bit binary output is converted to BCD using a custom-built binary-to-BCD converter. The converter was developed after examining all the possible output combinations of the multiplier in Logisim and designing the final logic circuit in Proteus. Comparator circuit (74LS85) was also added to verify overflow. If the result exceeds 99, a flag is produced to indicate that the output is now invalid in BCD. All operation modules run in parallel and make use of the same inputs from the registers.

e. Multiplexer (MUX)

Four 4-bit multiplexers (e.g., 74153) are used to drive the corresponding result of adder, subtractor, or multiplier to output. Two are for the units place of the result and two are for the tens place. Dividing the result in this way gives full 2-digit BCD display and is flexible to choose as per operation. Selector switches indicate through which operation output is driven to the registers.

The operation to be performed is selected through a 2-bit selector input:

- 00 – Addition
- 10 – Subtraction
- 01 – Multiplication

f. Output registers

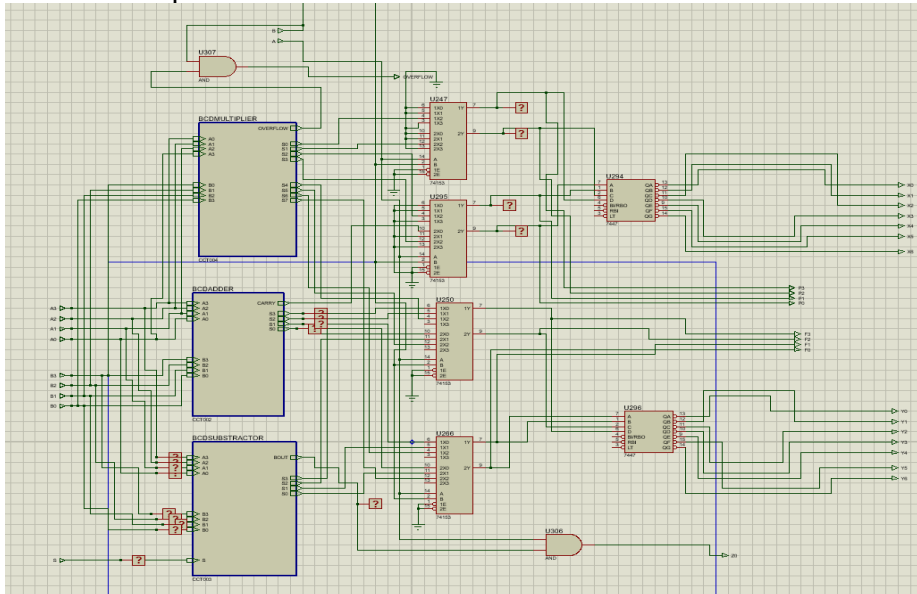
Two more 74173 ICs are utilized to store the result of the multiplexers. The registers are clock-driven and provide clean, stable output to the 7-segment display system. Storing the result, the display does not blink nor change abruptly in transitions.

g. Display System

The output of the BCD from the output registers is supplied to 7447 BCD-to-7-segment decoder ICs, which are again wired to common anode 7-segment displays for input and output values. A special common cathode 7-segment display is utilized to show the minus sign. This provides easy enablement of the sign display by driving only the appropriate segment (typically the central segment 'g') with additional logic control. The use of both anode and cathode types ensures reliable display of numeric values and signs.

4. Components and description

a. Mini processor



Components: BCD multiplier

BCD Adder

BCD subtractor

AND gate

U294

U296

Input: A3 A2 A1 A0

B3 B2 B1 B0

S (for subtractor)

A (selector)

B (selector)

Output: X0 X1 X2 X3 X4 X5 X6 (for output)

P0 P1 P2 P3 (for output registers)

F3 F2 F1 F0 (for output registers)

Y0 Y1 Y2 Y3 Y4 Y5 Y6 (for output)

Z0 (borrow out of the subtractor to display negative sign)

Circuit:

The mini-processor has three important arithmetic blocks: a BCD adder, a BCD subtractor, and a BCD multiplier, each with 4-bit BCD input A (A3–A0) and input B (B3–B0). Two selector switches A and B select between operations such that 00 selects addition, 10 selects subtraction, and 01 selects multiplication. Another one is using an S control input to the subtractor whereby 2's complement operation occurs when subtraction is chosen through

addition. To the subtractor, 1 is added to the inverted B input and furnished. Outputs of the arithmetic units are furnished to two 74153 multiplexers, which choose the right output for tens and units digit based on the operation. The multiplier produces a 7-bit binary output (X0–X6), which is translated into BCD by a special logic circuit. Outputs are represented by a 7447 BCD-to-7-segment decoder driving common anode 7-segment displays. A negative sign is conditionally shown in subtraction with an AND gate between the borrow output and selector A, so only the sign is shown for negative outputs of subtraction. The following module design has proper BCD arithmetic operations with neat presentation and logic flow control.

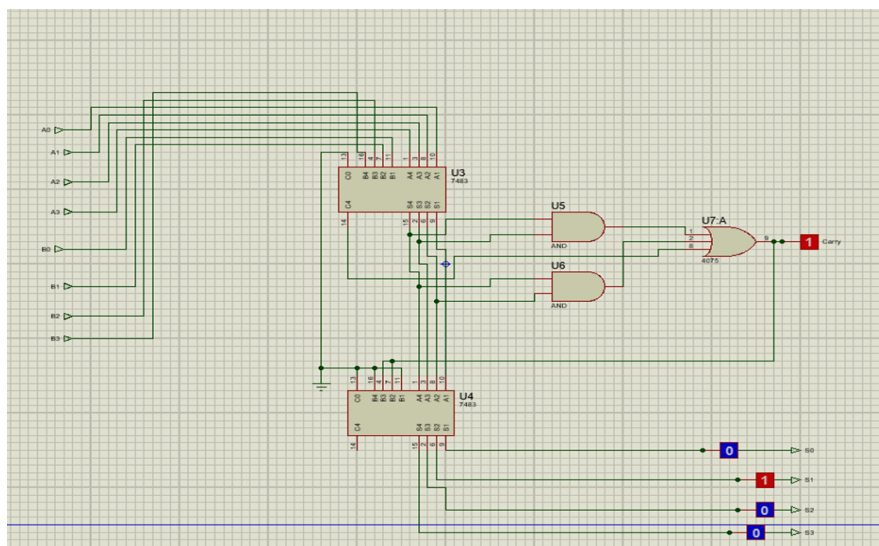
b. BCD adder

Components: 7483 – 4 bit binary adder

Inputs: A0, A1, A2, A3

Outputs: B0, B1, B2, B3

Circuit:



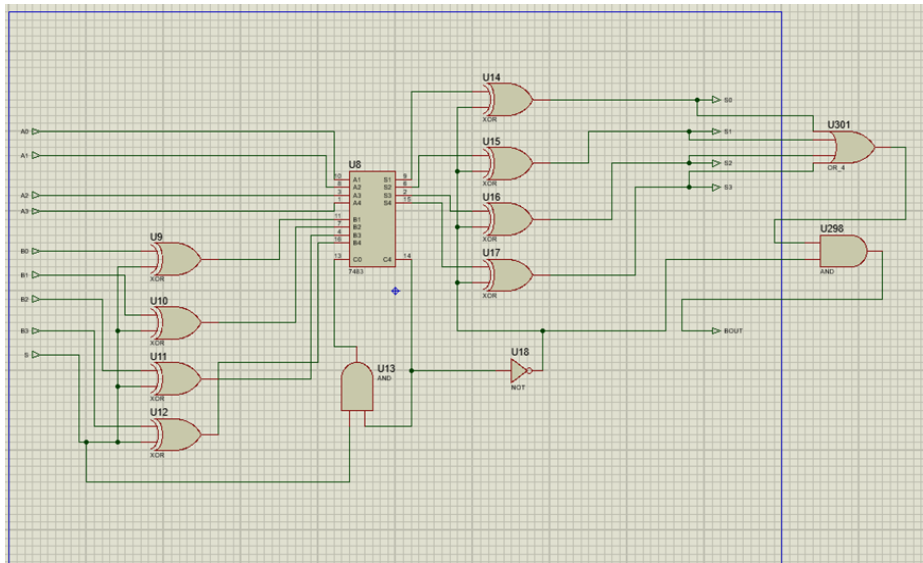
The BCD adder uses a 7483 4-bit binary adder to add two BCD inputs A (A3–A0) and B (B3–B0). The adder merely adds the binary value together, and if the sum exceeds 9 (1001 in binary) or produce a carry-out, a correction value of 6 (0110) is added by another adder stage or logic circuit to place the result in valid BCD form. This correction serves to translate invalid binary digits like 1010 (10) or more to legitimate decimal numbers. The adder yields a 4-bit BCD units digit result (P0–P3) and, if needed, a carry is passed on to the tens digit circuitry (F0–F3) for creating a compound two-digit result. The resultant output is routed via the 74153 multiplexer based on the operation selector (A = 0, B = 0 for addition). Since the adder does not contain any borrow logic, the logic for the minus sign is automatically suppressed during addition to give clear and precise display output

c. BCD subtractor

Components :7483 -4 bit binary adder

Inputs: A0 A1 A2 A3

Circuit:



The BCD subtractor is realized using a 7483 4-bit binary adder operating in the 2's complement mode of subtraction. When it is operating in the mode of subtraction ($A = 1, B = 0$), input B (B_3-B_0) is bit-by-bit inverted by XOR gates governed by the S input, and a logical '1' is fed into the carry-in pin of the adder, thus performing $A - B$. If the output produces a bigger binary number than 9 or an erroneous BCD digit, there is correcting logic that adds 6 (0110) and validates the output to BCD. The subtractor gives the tens (F_0-F_3) and the units (P_0-P_3) digits that get routed through multiplexers. Borrow condition is detected from the carry-out and used in conjunction with selector A through an AND gate to allow the display of the minus sign so that the minus sign would be present only for correct negative outputs in case of being in subtraction mode. The logic described above prevents display glitches such as the display of "-0" and allows the subtractor being under BCD constraints.

d. BCD multiplier

Components: Binary multiplier
Binary to BCD converter
comparator

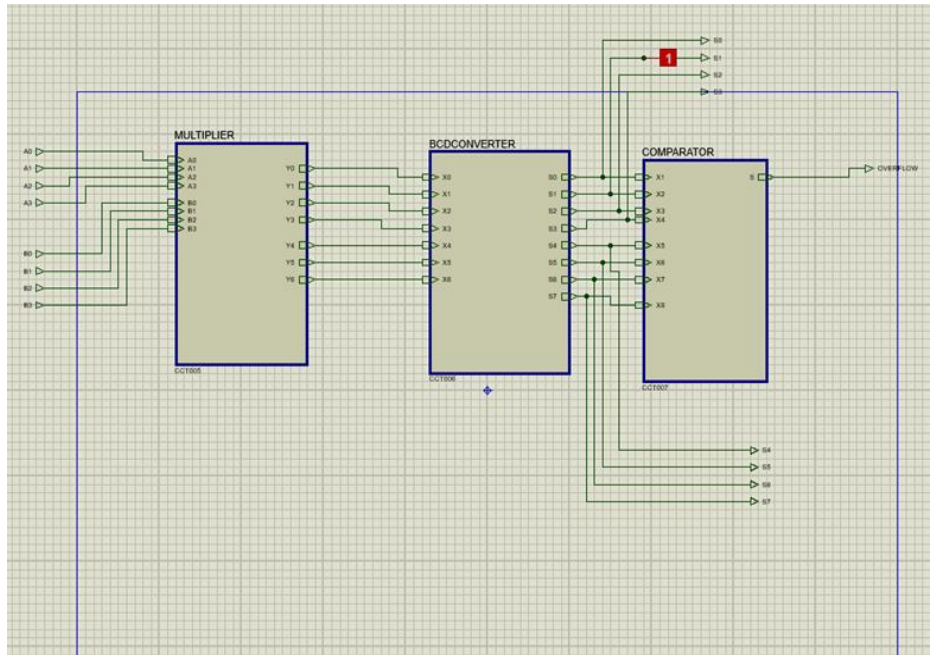
Inputs:

- A [A0, A1, A2, A3]
- B [B0, B1, B2, B3]

Outputs:

- S [S0, S1, S2, S3, S4, S0]

Circuit



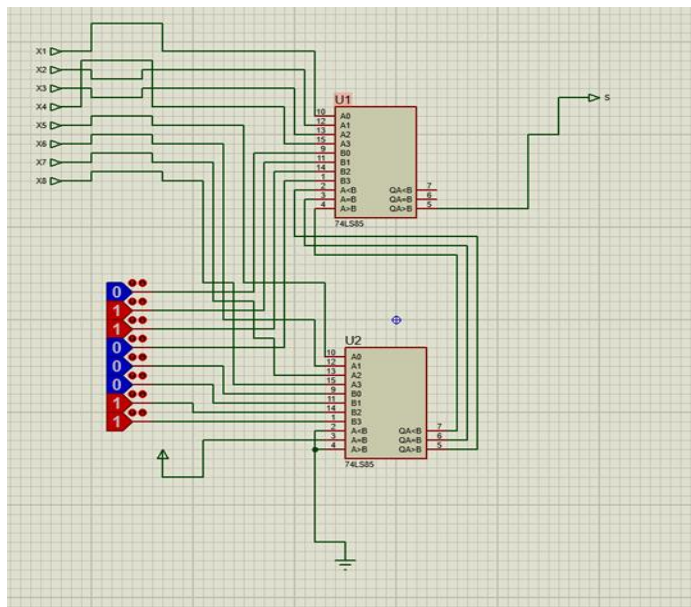
Comparator

Components: 74LS85

Input: X1 X2 X3 X4 X5 X6 X7 X8

Output: s (if results are more than 99 flag will indicate 1 and otherwise 0)

Circuit:



The comparator is used to detect overflow while multiplying so that the outcome remains in valid BCD range from 0 to 99. For comparing binary output of the multiplier with a fixed threshold of decimal 99 (binary 1100011), a 74LS85 4-bit magnitude comparator is used.

Since the multiplier has the capability to generate outputs up to 81 (and theoretically beyond for garbage inputs), the comparator is employed to identify if the 7-bit binary output is over this upper BCD limit. If the output is greater than 99, the comparator produces a high signal which is utilized for setting an overflow flag. Such a flag may be used to alert the user that the resulting display in the BCD context is not valid, for instance, by preventing the display or by displaying a visual warning. The inclusion of the comparator makes the system more accurate and reliable through valid BCD output guaranteeing.

e. Binary to BCD converter

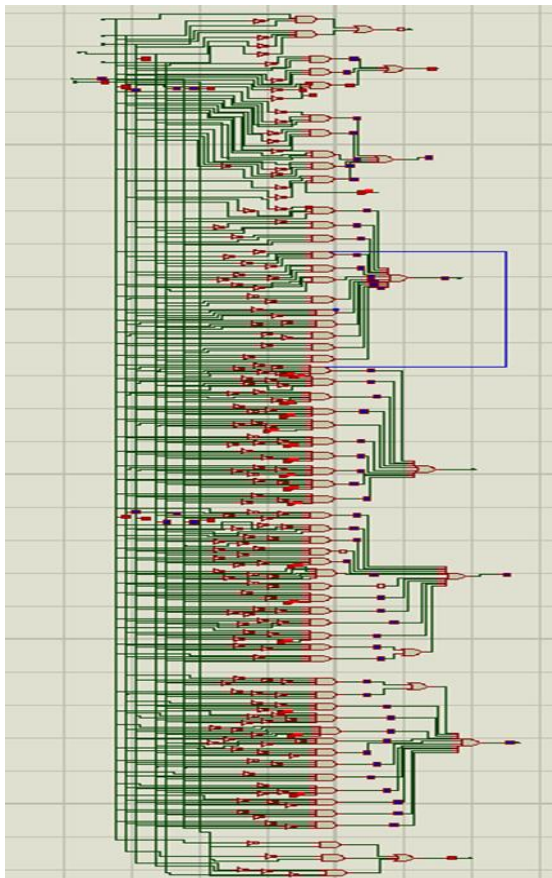
Components: AND gate

OR gate

Input: X0 X1 X2 X3 X4 X6

Output: S0 S1 S2 S3 S4 S5 S6 S7

Circuit:



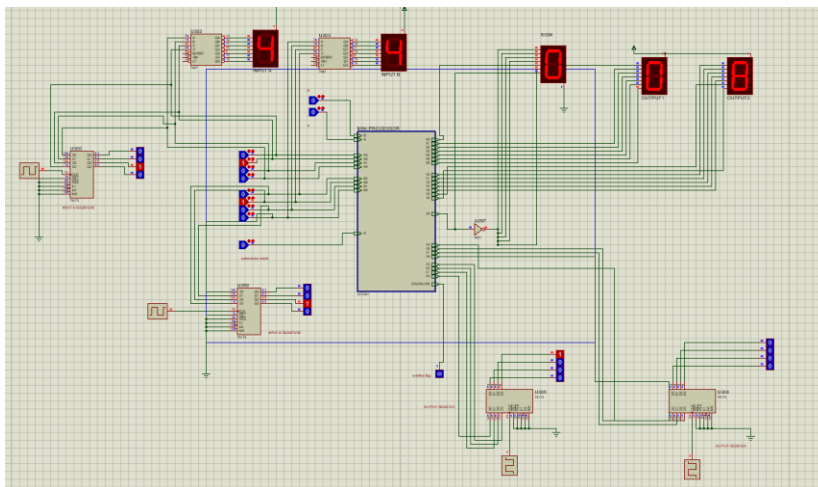
Binary-to-BCD converter is a one-off logic circuit used to convert the 7-bit binary output of the multiplier into the correct BCD form. Since special ICs for this were not available in Proteus, the converter was manually simulated on the truth-table level. The different possible binary outputs (0–81) of the multiplier were initially simulated and tabulated in

Logisim and their corresponding BCD representations determined. Two independent 4-bit BCD outputs were thus produced by combinational logic circuits: F0–F3 for the tens digit and P0–P3 for the units digit. They were sent to the display system through multiplexers. The converter also properly represents binary numbers like 0101001 (41 in decimal) as 0100 and 0001 (BCD 4 and 1). This implementation is compatible with binary multiplication results to be shown by standard BCD display logic and is fully compatible with the 7447 7-segment decoder.

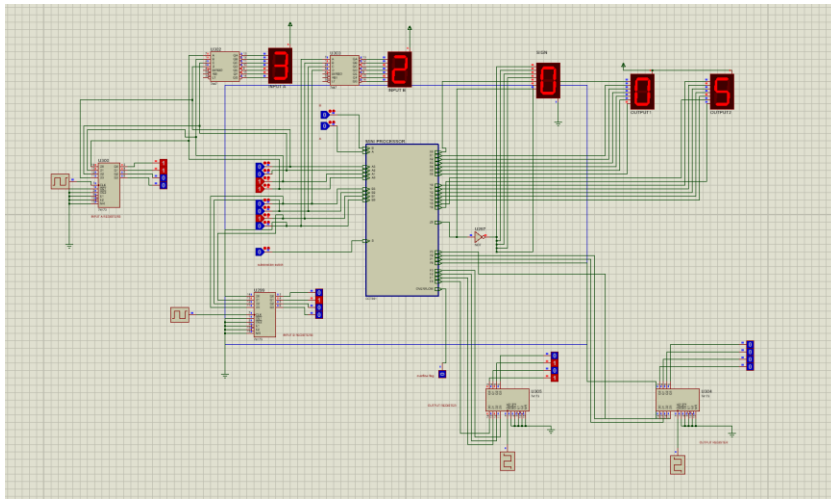
5. Simulation results

Addition

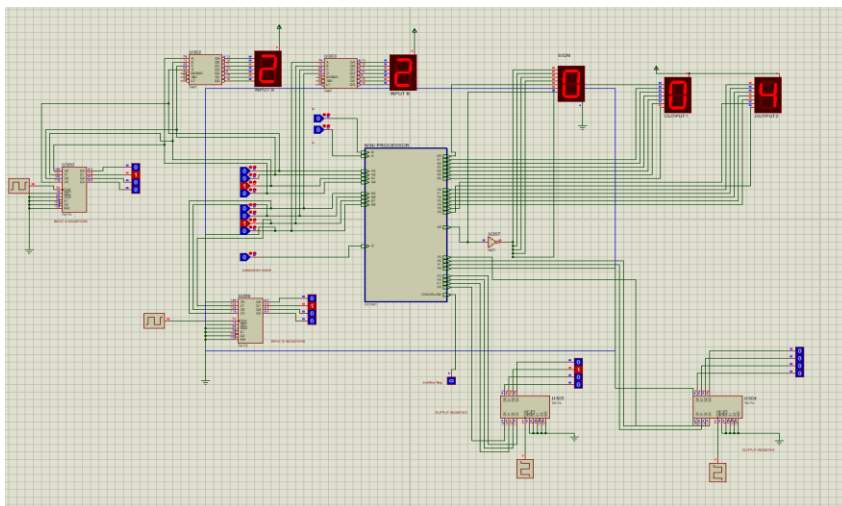
1. $4+4=8$



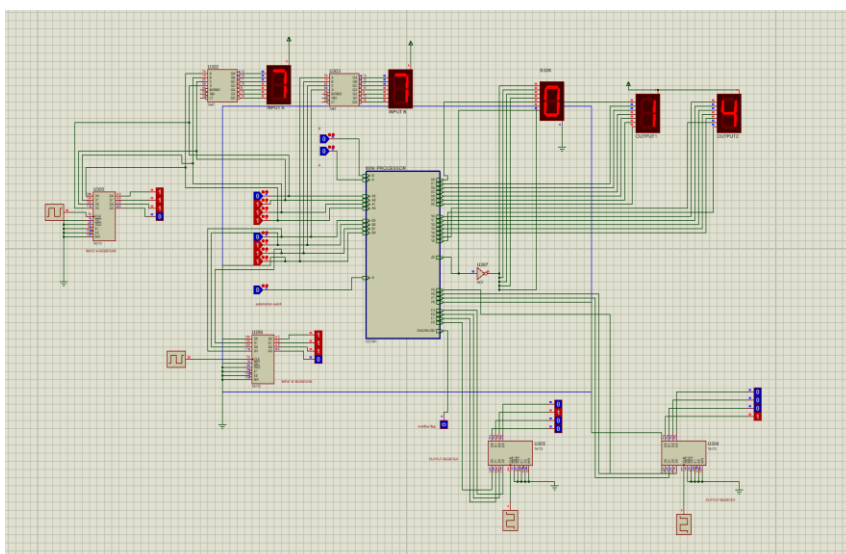
2. $3+2=5$



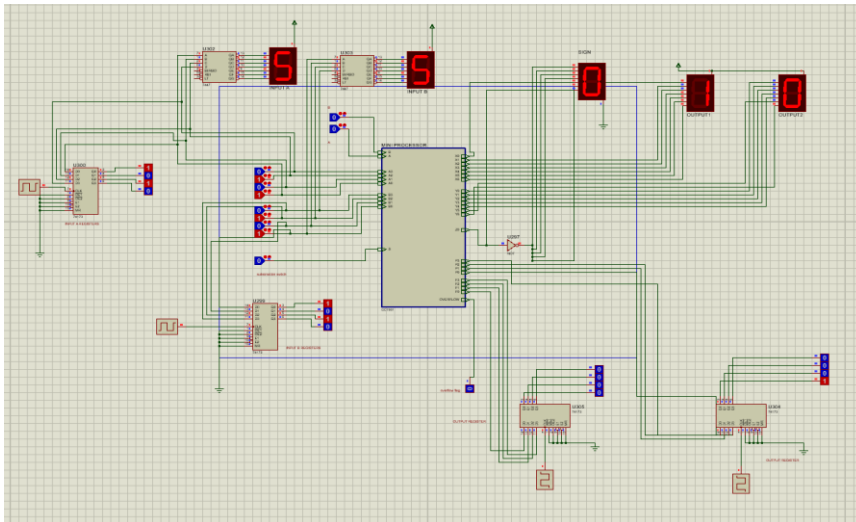
3. $2+2=4$



4. $7+7=14$

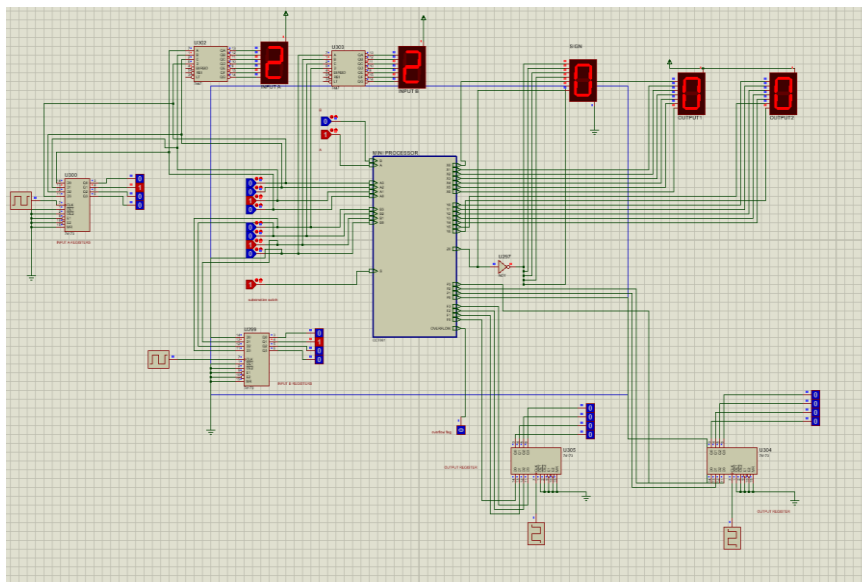


5. $5+5=10$

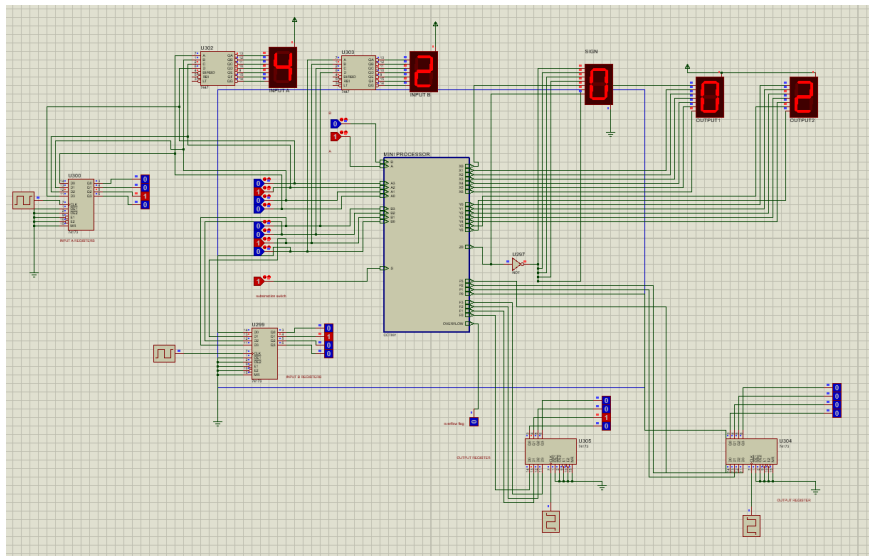


Subtraction

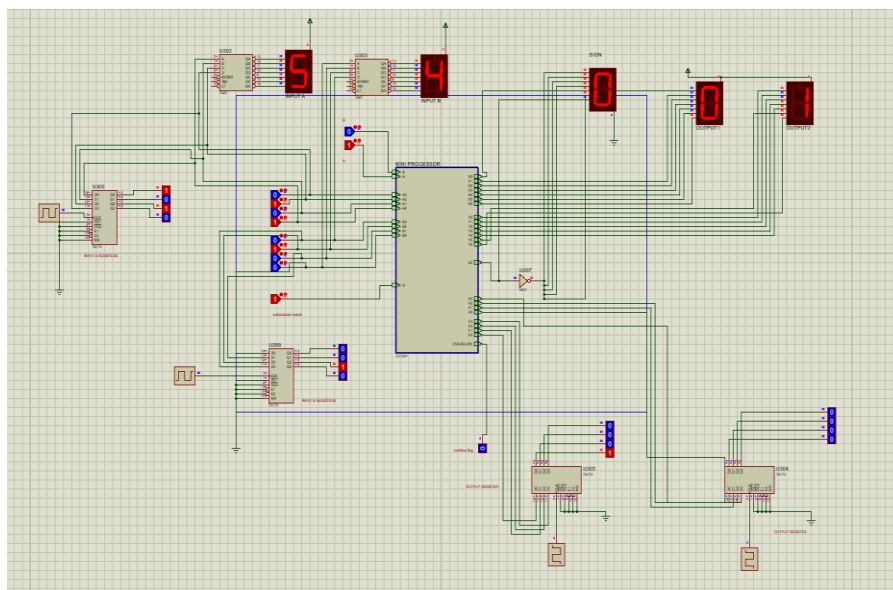
1. $2-2=0$



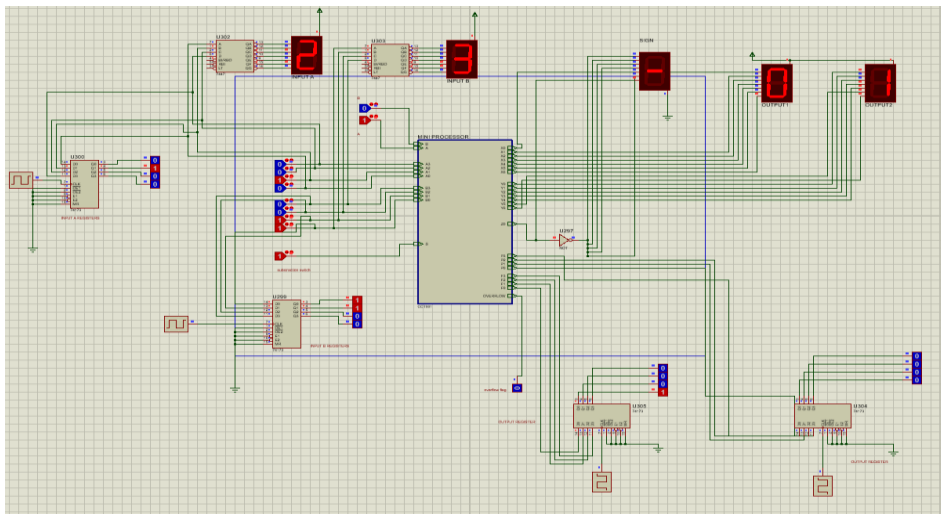
2. $4-2=2$



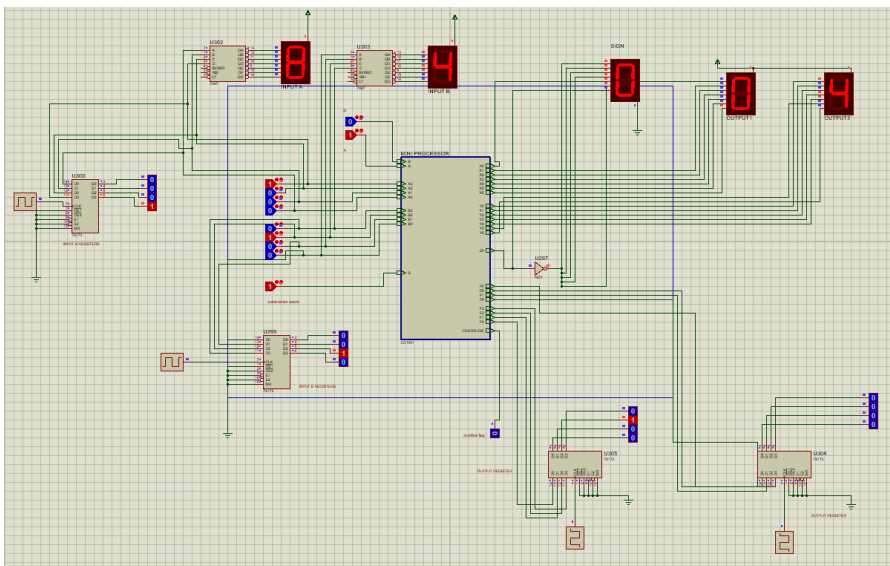
3. $5-4=1$



4. $2-3 = -1$

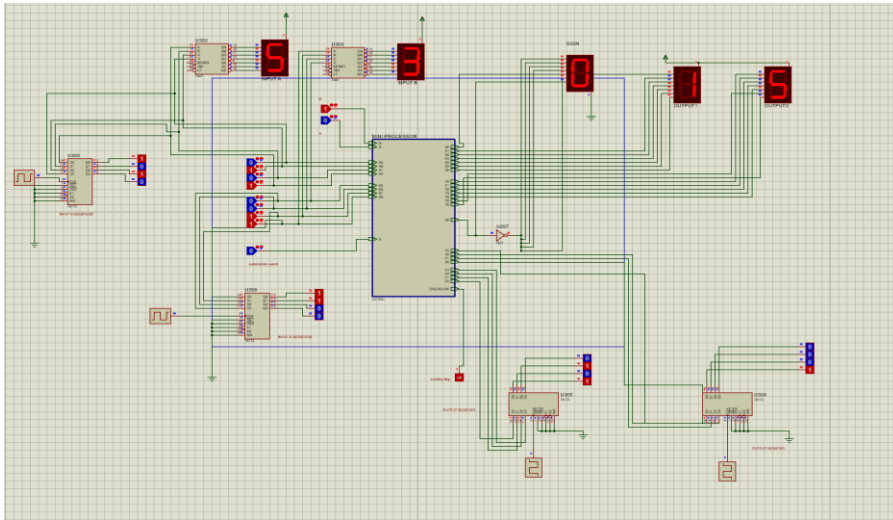


5. $8 \cdot 4 = 4$

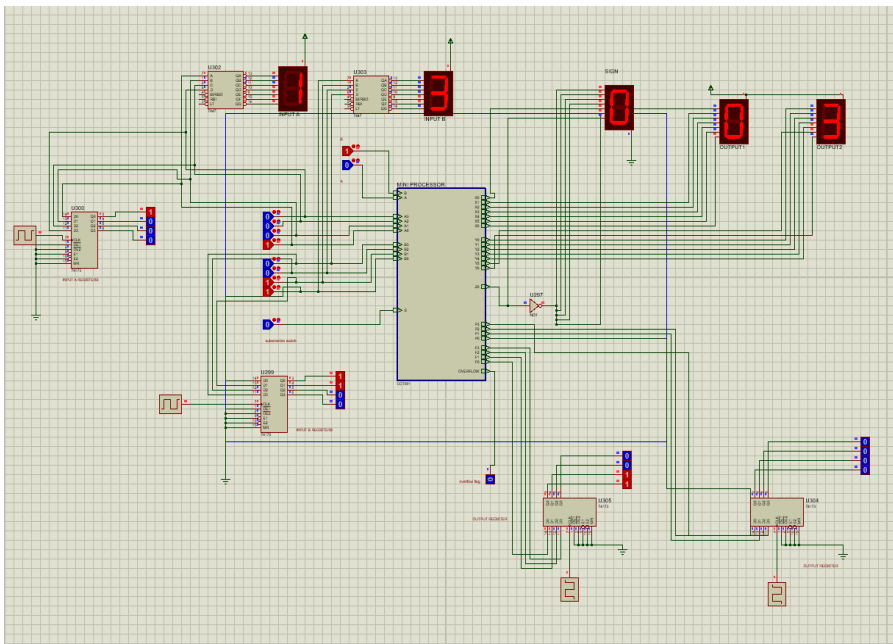


Multiplication

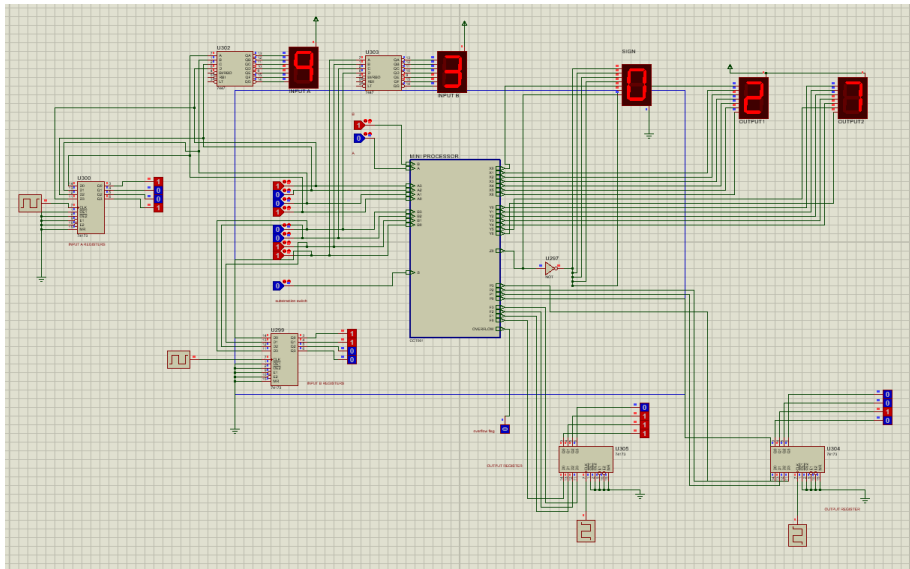
1. $5 \times 3 = 15$



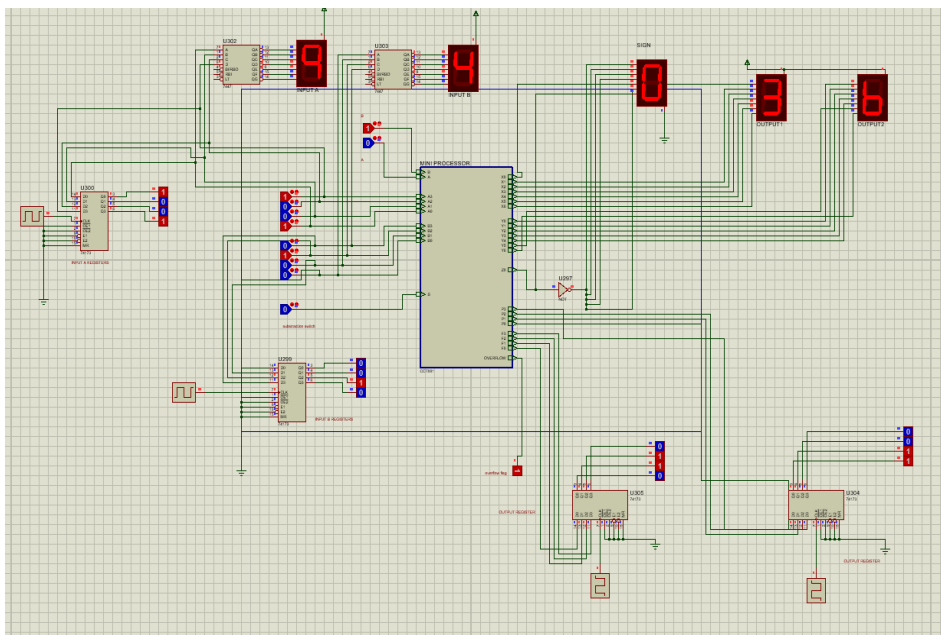
2. $1 \times 3 = 3$



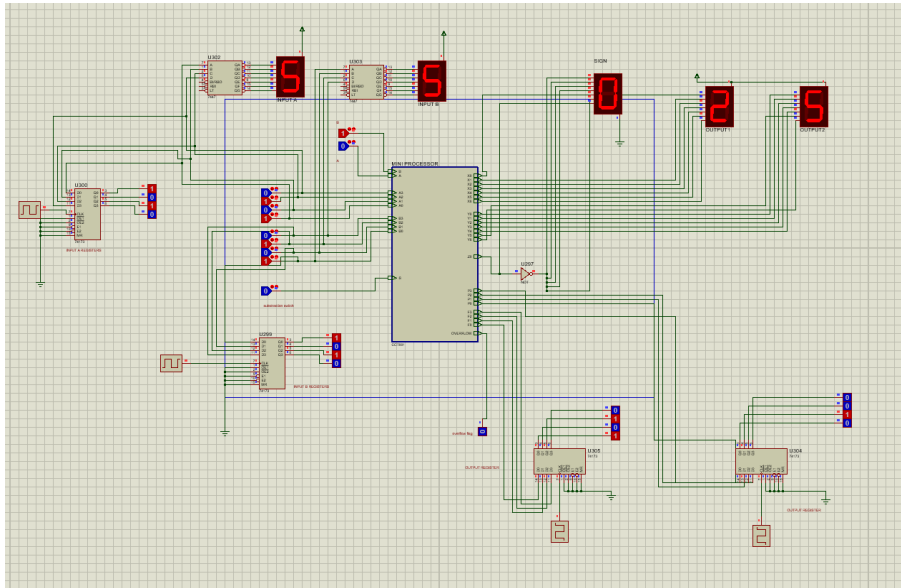
3. $9 \times 3 = 27$



4. $9 \times 4 = 36$



5. $5 \times 5 = 25$



6. Challenges and fixes

1. BCD Correction Logic for Addition

Problem: Original BCD adder generated binary results > 9 without correction.

Solution: Added conditional tests to check whether result was > 1001 or carry-out was 1.

Added 0110 (decimal 6) with an extra adder to shift the result into valid BCD range.

Learning: Learned why BCD correction is needed for non-binary-compliant results.

2. Minus Sign Displayed on $0 - 0$

Problem: Minus sign was being shown even on subtracting $0 - 0$.

Solution: Added an AND gate between borrow signal and a check for non-zero result. Now the minus sign is shown only when the result is negative but not zero.

Learning: Learned the requirement for conditional sign display logic for proper result.

3. Binary to BCD Conversion

Problem: No pre-programmed IC available in Proteus for 7-bit binary to BCD conversion.

Done manually by emulating all possible output combinations via truth tables within Logisim and then constructing the circuit within Proteus.

Mastered logic design and learned the art of converting truth-table logic to hardware.

4. Simulation Won't Start with Subtract Mode ON

Problem: The simulator would either generate incorrect results or hang whenever a simulation run was attempted with the Subtract selector (S) set to 1.

Solution: Set S to 0 at power-up and changed it after the system stabilized.

Learning: The behavior of startup circuits and initialization levels were understood through Proteus.

5. Register and Clock Signal Problems

Problem: The registers were not latching the values properly.

Solution: Checked the correct clock and control signals were being provided to all 74173 and 4042 ICs. Clean clock source with enable lines was used in synch.

Learning: Learned to use timing and control signals more effectively for correct output.

6. Multiplier Overflow Detection

Problem: Output > 99 is invalid in BCD but earlier there was no overflow indication.

Solution: Incorporated a 74LS85 comparator to detect output > 99 and set an overflow flag.

Lesson Learned: Recognize that a number needs to be checked in the arithmetic logic.

7. Displaying Negative Sign On Addition

- Problem: The negative sign kept coming up even for addition, especially the addition whose result came out as 0 or small numbers.

- Cause: The minus only showed when the borrow out line made it do so without checking what operation was being chosen.

- Fix: The AND gate was added between the borrow output and operation selector line A. To the extent that $A = 1$ signifies subtraction mode, minus is now only displayed when both a borrow is generated and subtraction is selected.

- Learning: The above has actually highlighted the importance of gating control signals in avoiding cross-interference during arithmetic operations and their output signals.

6. Conclusion

The project can design and simulate an addition, subtraction, and multiplication BCD arithmetic unit based on digital logic building blocks. With the inclusion of a BCD adder, BCD subtractor, and a custom-designed BCD multiplier-with ancillary items such as multiplexers, registers, binary-to-BCD converters, and overflow detection-the mini-processor can accept 4-bit BCD inputs and compute and display correctly valid outputs. Selector-driven multiplexers offer transparent switch-over among operations, and 7447 BCD-to-7-segment decoder-driven display system offers clear visible output. Problems such as minus sign display in an incorrect way, startup situations, and restraints in conversions were resolved by applying logical gating, initializations, and particularly designed circuits. The whole project improved the acquaintance with BCD logic, arithmetic circuits, system integration in digital, and simulation practices using software like Proteus and Logisim. The outcome is a working, instructional BCD arithmetic unit demonstrating practical application of theoretical ideas in digital design.

7. References

- [1] E. Babbage, BCD Adder Explained, YouTube. [Online]. Available: <https://youtu.be/z6HFYyQv4Es?si=fh0WUim0ebTbnaho>
- [2] A. Logic, BCD Subtractor Using 7483 IC, YouTube. [Online]. Available: <https://youtu.be/nV6s5Z8PHeY?si=3rDeNRMkEv2VMYIH>
- [3] J. Digital, Binary to BCD Converter Logic Circuit, YouTube. [Online]. Available: https://youtu.be/2QExUjO4GdQ?si=CEi_TTSvvppxEq1j
- [4] M. Ercegovac and T. Lang, "Binary-coded decimal digit multipliers," Proceedings of the 12th IEEE Symposium on Computer Arithmetic, pp. 66–72, 1995. [Online]. Available: https://www.researchgate.net/publication/3480208_Binary-coded_decimal_digit_multipliers
- [5] N. Kaur and M. Sharma, "Block diagram of 4×4 BIT multiplier working process," ResearchGate, [Online]. Available: https://www.researchgate.net/figure/Block-diagram-of-44-BIT-multiplier-working-process_fig1_344615501